Inventory analysis evaluates a business's stock levels, movement, and performance to optimize inventory management. It helps determine the right amount of stock to hold for meeting demand while minimizing costs like storage and obsolescence. Businesses use it to make data-driven decisions on purchasing, sales, and supply chain efficiency.

Core Purpose:

The primary goal is balancing stock to avoid overstocking, which ties up capital, and stockouts, which lose sales. Key metrics include inventory turnover (sales divided by average inventory), days of inventory on hand, and reorder points. This process identifies trends in fast- versus slow-moving items for better cash flow and profitability.

Common Methods:

ABC Analysis: Categorizes items by value—A (high-value, tight control), B (medium), C (low-value, minimal monitoring).

Inventory Turnover Ratio: Measures how often stock sells and replenishes, calculated as cost of goods sold divided by average inventory.

VED Analysis: Classifies by criticality—Vital (must-have), Essential, Desirable—for industries like healthcare.

Key Benefits:

Reduces carrying costs and waste from excess or obsolete stock.

Improves customer satisfaction by ensuring product availability.

Enhances forecasting and supplier negotiations through performance insights.

- Inventory analysis improves cash flow by optimizing stock levels to free up tied-up capital, reduce holding costs, and accelerate inventory turnover. It identifies inefficiencies like excess slow-moving stock or shortages that lead to lost sales opportunities. This data-driven approach ensures businesses hold just enough inventory to meet demand without unnecessary expenses.
- **Reduces Holding Costs:** Overstocking locks cash in unsold goods, incurring storage, insurance, and obsolescence costs. Analysis tools calculate optimal reorder points and safety stock, minimizing excess inventory by 20-30% in many cases. Faster turnover means quicker conversion of stock back to cash.
- **Boosts Inventory Turnover:** Key metrics like turnover ratio (COGS / average inventory) reveal slow sellers for discounting or discontinuation. Businesses achieve higher ratios through precise forecasting, turning inventory 4-6 times yearly instead of 2-3. This shortens cash conversion cycles and improves liquidity.
- **Prevents Stockouts and Lost Sales:** Analysis predicts demand to avoid shortages that halt sales and damage revenue. ABC analysis prioritizes high-value items (A category), ensuring availability for 80% of profits from 20% of stock. Reliable supply maintains steady inflows, stabilizing cash flow

Inventory analysis methods like ABC, XYZ, and EOQ classify and optimize stock to control costs, prioritize items, and determine ideal order quantities. ABC focuses on value-based categorization, XYZ on demand variability, and EOQ on cost-minimizing calculations. These techniques help businesses allocate resources efficiently and reduce waste.

- **ABC Analysis:**

ABC analysis divides inventory into three categories based on annual consumption value using the 80/20 Pareto principle. A items (20% of stock, 80% value) get tight controls like frequent reviews; B items (30% stock, 15% value) moderate oversight; C items (50% stock, 5% value) minimal monitoring. This prioritizes high-impact items for better cash flow and efficiency.

- **XYZ Analysis:**

XYZ analysis categorizes by demand predictability, complementing ABC in a 9-cell matrix (e.g., AX, CZ). X items have stable demand (low variability, CV <10%); Y moderate (CV 10-30%); Z erratic (CV >30%). It sets safety stock levels—low for AX (predictable high-value), high for AZ/CZ (volatile)—to match forecasting accuracy.

- **EOQ Model:**

Economic Order Quantity (EOQ) calculates optimal order size to minimize total costs:
EOQ=2DSH*EOQ*=*H*2*DS*
where D is annual demand, S ordering cost per order, H holding cost per unit. It balances ordering frequency against storage expenses, ideal for stable-demand items, preventing overstocking.